

Integrating MATSim and Microscopic Railway Simulation for Passenger Transport Digital Twinning

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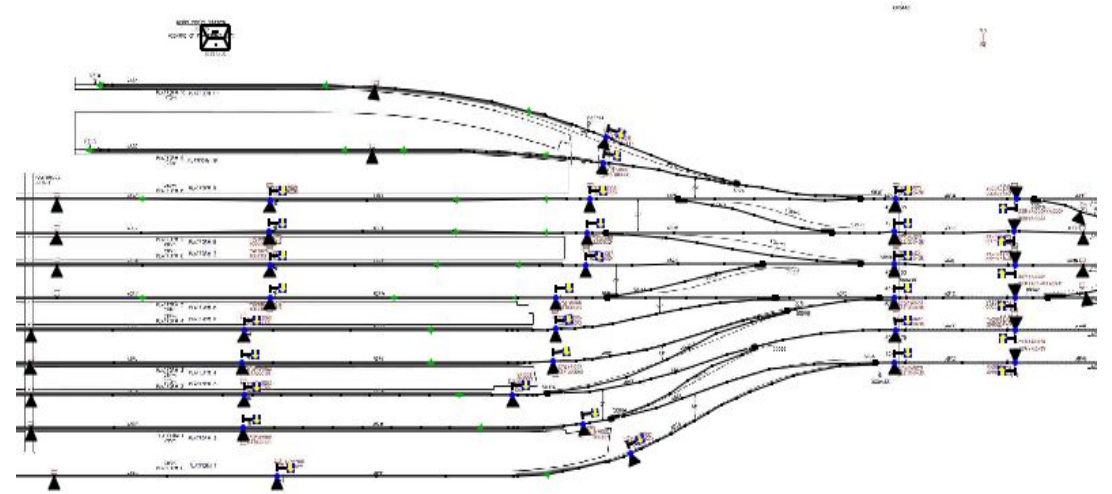
MATSim User Group Meeting 2026
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Why?

- MATSim captures large-scale passenger behaviour and multimodal interactions.
- Railway operations involve fine-grained dynamic constraints:
 - Signalling and Interlocking
 - Platform and route constraints
 - Train headways
 - Delay propagation
- Operational effects can influence:
 - Journey times
 - Reliability
 - Passenger mode choices

BRaVE

- Birmingham Railway Virtual Environment
 - Simulation of a railway network.
 - including train movements, signal interlocking, signaller and driver behaviour.
 - Core simulator connects to other systems.
 - Radio Network simulator, Power network simulator, Passenger simulator, External control systems, Post processing analysis.
 - Editors
 - Track, rolling stock, signalling, timetable
 - Editors / Importers.

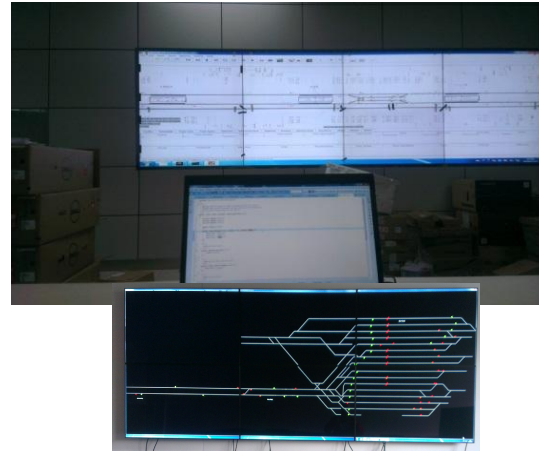


BRaVE

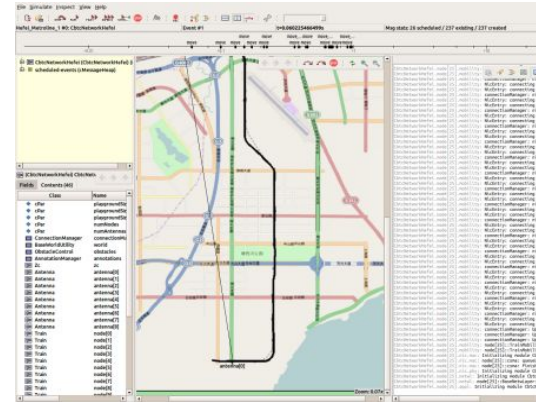
Train control



Signalling



Comms



Power



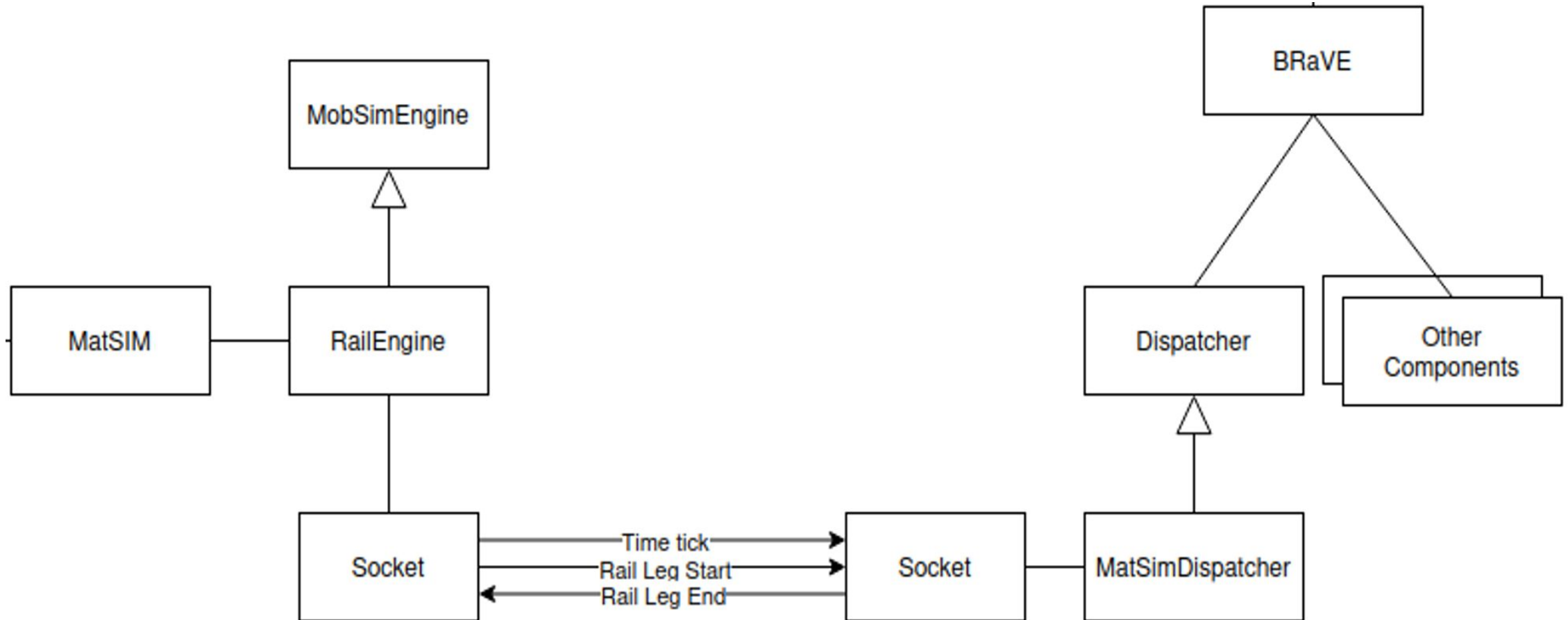
Why?

MATSim Capability	BRaVE Capability
Agent travel demand	Train movement dynamics
Route and mode choice	Signalling and Interlocking
Departure time choice	Dispatching and conflict resolution
Multimodal trip construction	Infrastructure & capacity constraints
Passenger behaviour	Delay propagation and reliability

What?

1. MATSim initiates rail leg
2. Passenger is transferred to BRaVE
3. BRaVE determines feasible train/service
4. Train operates under microscopic constraints
5. Passenger arrives at destination
6. Passenger returns to MATSim
7. MATSim evaluates the resulting journey

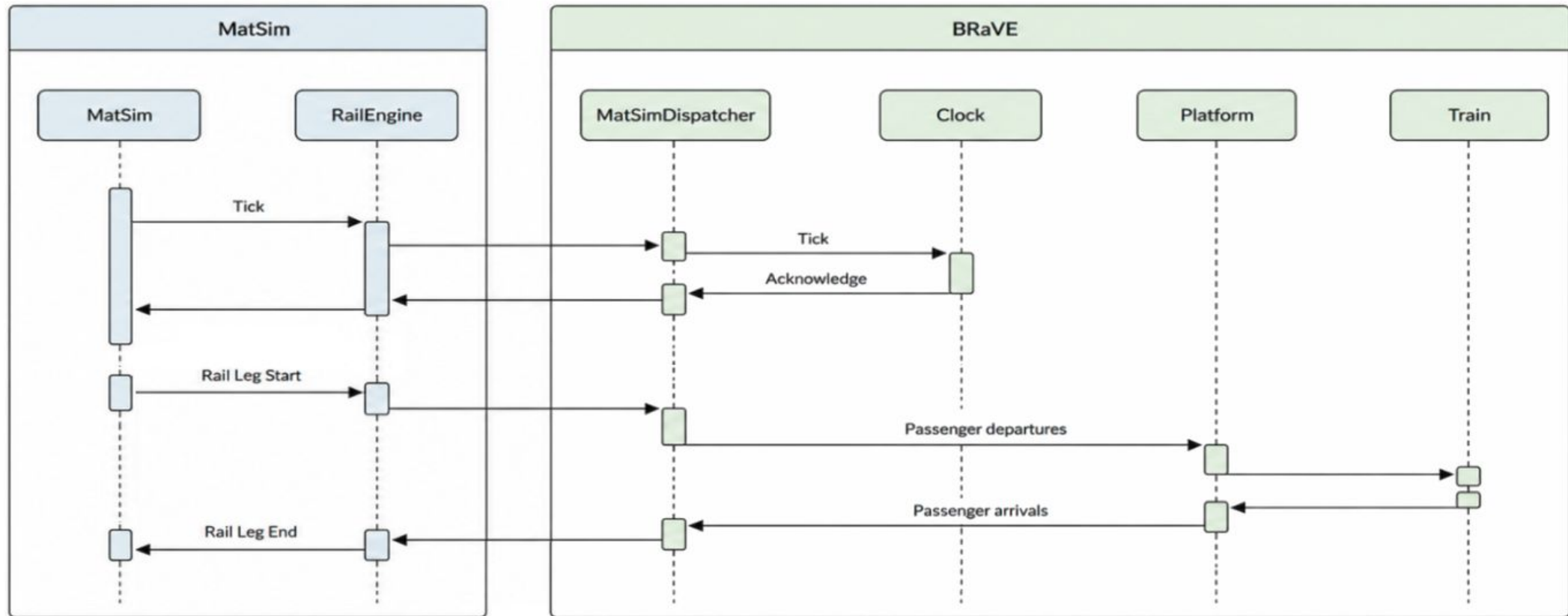
How?



How?

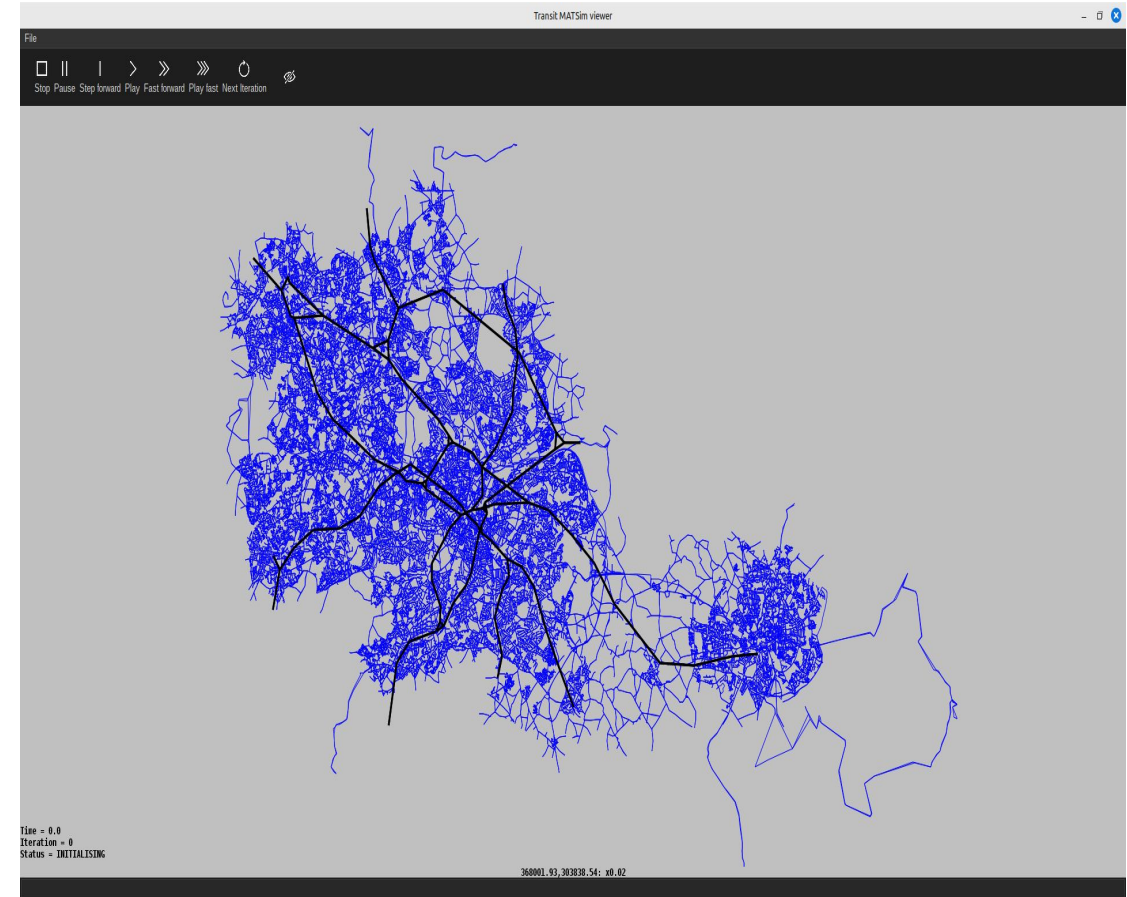
- Time synchronisation
 - MATSim time ticks are intercepted, blocked and used as clock for BRaVE.
- Rail-Leg interception
 - On start of Rail Leg, legs are passed to BRaVE.
 - BRaVE handles platform selection, boarding, alighting at destination and returns Rail Leg to MATSim for completion.

MATSim/BRaVE Sequence



CLD 2 West Midlands Passengers Rail and Road

- Capacity
 - Can infrastructure absorb rail demand?
 - Where do bottlenecks emerge?
- Investment
 - How do changes affect passenger behaviour
 - What are the implications for mode shift and emissions
- Reliability
 - How does rail performance affect mode choices?
 - How does rail disruption propagate to passengers?



Next steps

- Generalise the message interface
 - Move from simulator-specific messages to a generic, extensible message format.
- Move towards a federation architecture.
 - Use Portico/HLA as the federation layer.
- Federate additional simulators
 - MATSim for passenger behaviour and mode choice.
 - BRaVE for railway operations.
 - SUMO, Airport model, Port model.